

Telugu Handwritten Character Segmentation Using YOLOv11

A Project Report submitted in partial fulfillment of the requirements
for the award of the degree of

BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE AND ENGINEERING

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Hyderabad

2026

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DECLARATION

I hereby declare that the project report entitled **Telugu Handwritten Character Segmentation Using YOLOv11**. is an original work done in the Department of Computer Science and Engineering, GITAM School of CSE, GITAM (Deemed to be University) submitted in partial fulfillment of the requirements for the award of B.Tech Degree in Computer Science and Engineering.

The work has not been submitted to any other college or University for the award of any degree or diploma.

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CERTIFICATE

This is to certify that the project report entitled “**Telugu Handwritten Character Segmentation Using YOLOv11**” is a Bonafide record of work carried out by **S ROHIT KUMAR HU22CSEN0102032, P SAI KRISHNA HU22CSEN0100302, K MOHAN KRISHNA HU22CSEN0100105, C GANESH SAI SANTOSH HU22CSEN0101777**, students of this department submitted in partial fulfillment of requirements for the award of B.Tech Degree in Computer Science and Engineering.

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I would also like to thank my friends who helped me to make my work more organized and well-stacked till the end.

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1. Abstract

Handwritten Telugu character segmentation is a challenging task due to the complex structure of the script, which includes curves, compound characters, and variations in handwriting styles. Traditional image processing methods often struggle to accurately segment such text, especially in the presence of noise and overlapping characters. To overcome these limitations, this project proposes a deep learning-based approach using the **YOLOv11** model for efficient character segmentation.

The system is trained on a custom annotated dataset prepared using **Label Studio**, where individual Telugu characters are labeled with bounding boxes. The implementation is carried out in a Python environment managed using **Anaconda**, ensuring smooth training and execution. The trained model is capable of processing images, videos, and real-time camera input to detect and localize characters with high accuracy.

Performance is evaluated using standard metrics such as precision, recall, and mean Average Precision (mAP), showing that the model achieves reliable results with fast processing speed. The proposed system provides an effective solution for handwritten Telugu text segmentation and can be applied in areas such as document digitization, handwriting analysis, and development of OCR systems for regional languages.

2. Introduction

Handwritten character recognition plays a crucial role in converting physical documents into digital formats, enabling efficient storage, retrieval, and analysis of textual data. Telugu, being a structurally rich and complex script with numerous curves, loops, and compound characters, presents significant challenges for accurate segmentation, particularly in handwritten form where variations in writing style are common.

This project focuses on developing a robust and efficient segmentation system using the **YOLOv11**, a state-of-the-art object detection model known for its speed and accuracy. The system is trained on a custom dataset annotated using **Label Studio**, ensuring precise localization of individual characters through bounding boxes. The implementation is carried out in a Python-based environment managed with **Anaconda**, enabling smooth training and deployment.

The proposed system supports multiple input formats, including static images, recorded videos, and real-time camera feeds, making it adaptable to various real-world applications. Emphasis is placed on achieving high segmentation accuracy along with real-time processing capability. The results demonstrate that the model effectively detects and segments handwritten Telugu characters, making it suitable for applications such as document digitization, handwriting analysis, and development of optical character recognition (OCR) systems.

A. Background

- Handwritten character recognition is a key area in **computer vision** and **pattern recognition**.
- It enables conversion of physical handwritten documents into **digital text formats**.
- Plays an important role in:
 - Document digitization
 - Data storage and retrieval
 - Automation of manual processes

B. Challenges in Telugu Handwritten Text

- Telugu script is **complex and highly structured**.
- Major challenges include:
 - Presence of **curves and loops**

- Use of **compound characters** (*vattulu*)
- Use of **modifiers** (*guninthalu*)
- **Variations in handwriting styles**
- **Overlapping and touching characters**
- **Inconsistent spacing** between characters

C. Importance of Segmentation

- Segmentation is a **preprocessing step** in OCR systems.
- It involves: Separating text into **individual characters**
- Accurate segmentation is important because:
 - It directly affects **recognition accuracy**
 - Errors in segmentation lead to **incorrect classification**

D. Deep Learning-Based Approach

- Recent advancements in deep learning provide better solutions.
- This project uses **YOLOv11** for segmentation.

Key Features of YOLOv11:

- Real-time object detection
- High accuracy
- Single-stage detection (fast processing)
- Detects multiple characters simultaneously

3. Literature Review

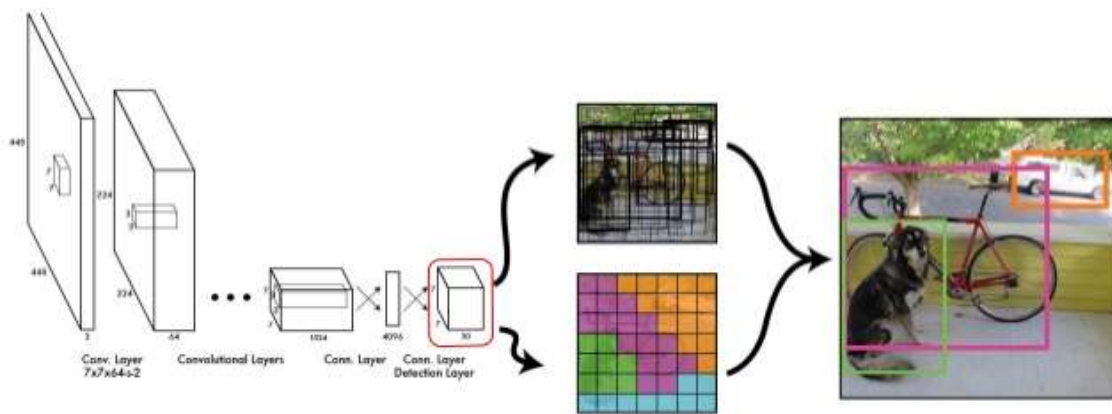
Earlier approaches for handwritten character segmentation relied on traditional image processing techniques such as thresholding and contour detection. These methods often fail when dealing with complex scripts like Telugu. With the advancement of deep learning, models like CNNs and YOLO have significantly improved object detection and segmentation tasks. YOLO-based models are particularly effective due to their speed and accuracy in real-time detection scenarios. This project utilizes YOLOv11 to overcome the limitations of traditional methods and provide efficient segmentation of handwritten Telugu characters.

4. Problem Identification & Objectives

Problem Identification

Segmenting handwritten Telugu characters is a challenging task due to the inherent complexity of the script. Telugu characters often consist of intricate curves, compound formations, and overlapping strokes, which make it difficult for traditional segmentation methods to accurately isolate individual characters. Additionally, variations in individual handwriting styles further increase the difficulty of achieving consistent and reliable segmentation. These challenges necessitate the use of advanced deep learning-based approaches for effective character detection and segmentation.

YOLO: You Only Look Once



Redmon et al. [You Only Look Once: Unified, Real-Time Object Detection](#), CVPR 2016

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Objectives

- Develop an efficient segmentation system using YOLOv11
- Create a well-annotated dataset using Label Studio
- Enable real-time detection of handwritten Telugu characters
- Improve segmentation accuracy and robustness
- Ensure the system can handle variations in handwriting styles

5. Existing System

Existing systems include:

- Traditional image processing methods
- Basic deep learning models

Limitations

- Poor performance on complex scripts
- Low accuracy in real-world handwritten data
- Limited real-time capabilities

Details of Existing System and their drawbacks

6. Proposed System

The proposed system uses:

- **YOLOv11** for object detection and segmentation
- **Label Studio** for dataset annotation
- **Anaconda** for environment management

Advantages

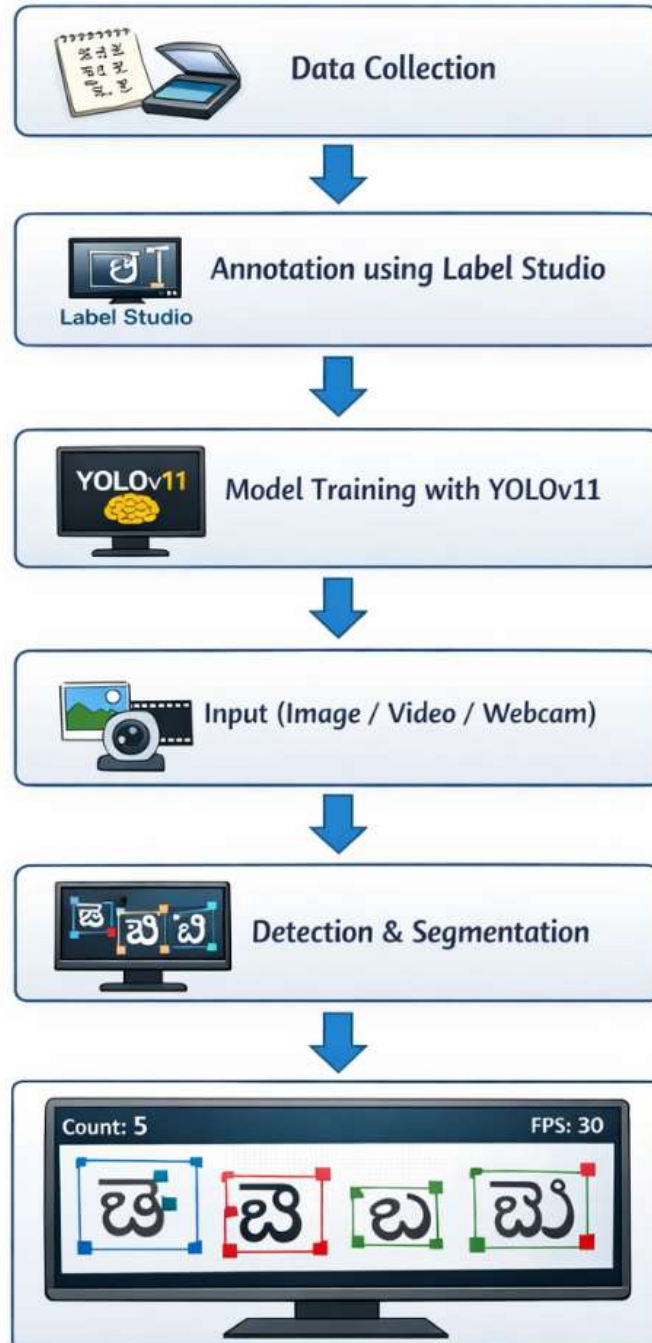
- High detection accuracy
- Real-time performance
- Flexible input handling (image, video, webcam)
- Scalable and efficient

Details of the Proposed System and how it improved over the existing system

7. System Architecture

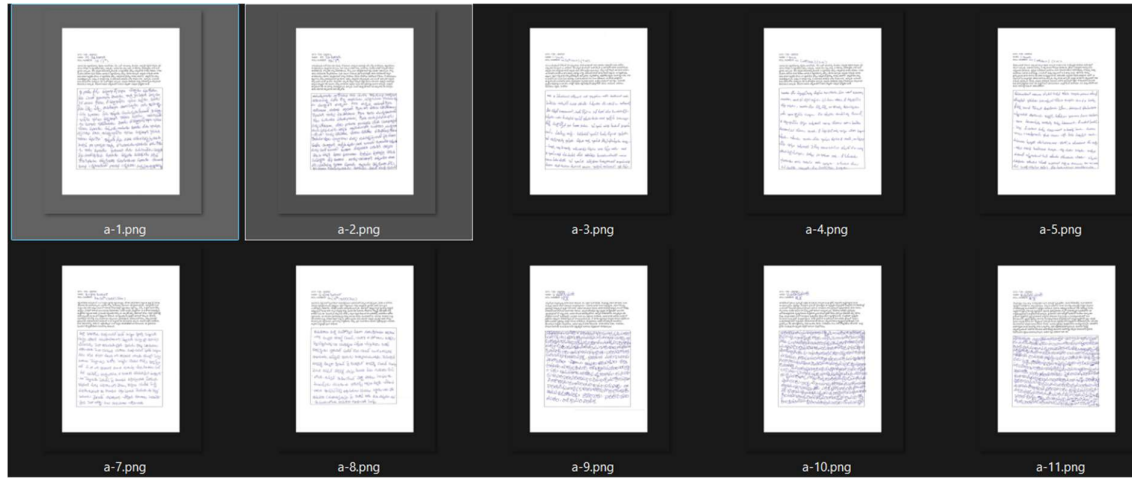
The system follows these steps:

Telugu Handwritten Character Segmentation System

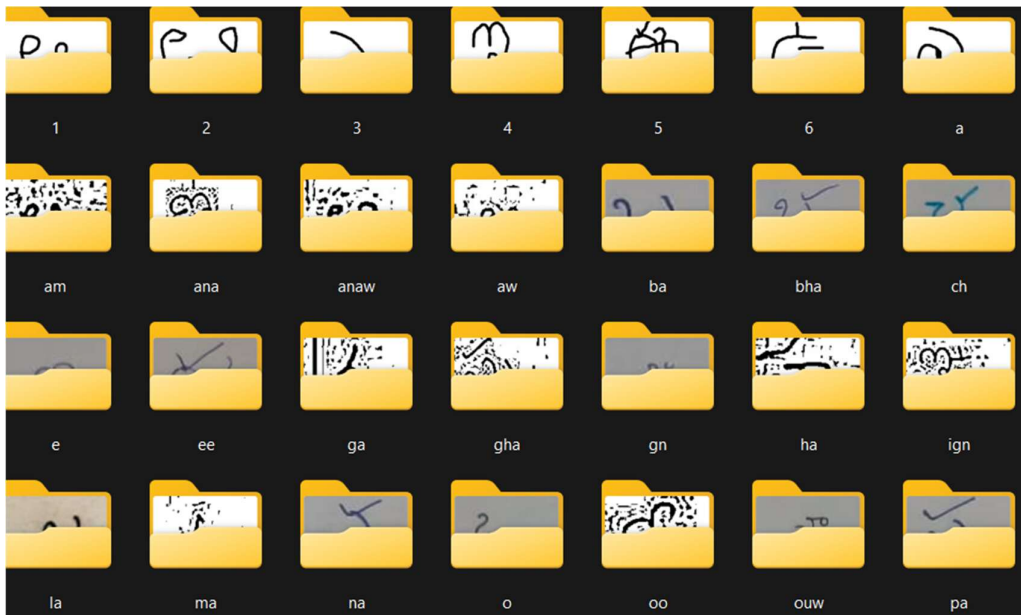


8. Process To Build

1. Data Collection



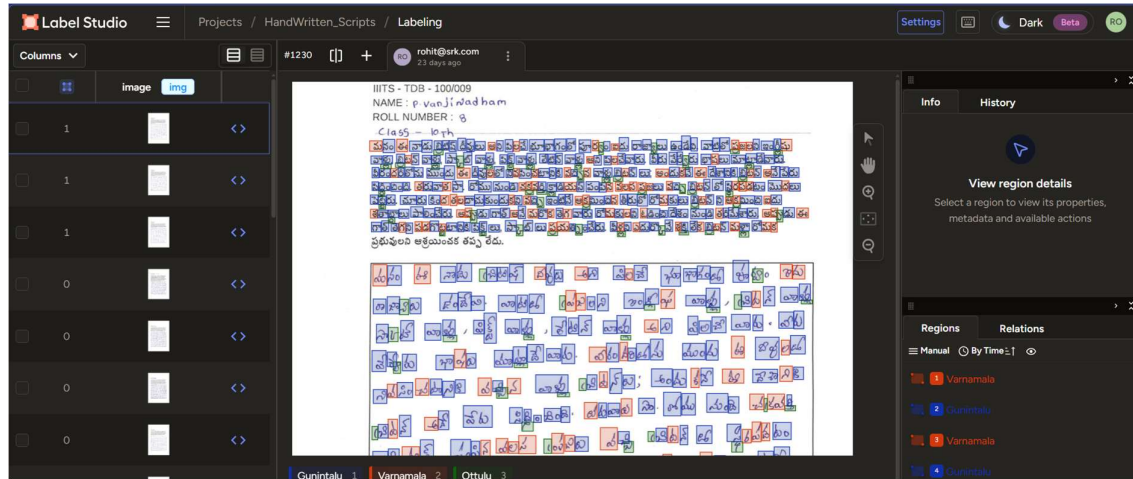
The first step involves collecting handwritten Telugu character data from various sources such as handwritten notes, scanned documents, and images. The dataset includes diverse writing styles to improve the generalization capability of the model. Proper data collection ensures that the model learns from real-world variations in handwriting.



2. Data Annotation using Label Studio



The collected images are annotated using Label Studio, where bounding boxes are drawn around individual Telugu characters. Each character is labeled appropriately to create a structured dataset. This annotation process is crucial for supervised learning, as it provides the ground truth required for training the YOLOv11 model.



3. Model Training using YOLOv11

The annotated dataset is used to train the YOLOv11 object detection model. During training, the model learns to identify and localize Telugu characters by predicting bounding boxes and class labels. The training process involves multiple iterations (epochs), optimization of weights, and evaluation using performance metrics.

```
!yolo detect train data=/content/data.yaml model=yolo11s.pt epochs=60 imgsz=640

Creating new Ultralytics Settings v0.0.6 file ✓
View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see https://docs.ultralytics.com
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11s.pt to 'yolo11s.pt': 100% 18.4MB
Ultralytics 8.4.23 Python-3.12.12 torch-2.10.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
engine/trainer: agnostic nms=False, amp=True, angle=1.0, augment=False, auto.augment-randaugment, batch=16, bgr=0.0, box=7.5, ca
Downloading https://ultralytics.com/assets/Arial.ttf to '/root/.config/Ultralytics/Arial.ttf': 100% 755.1KB 16.8MB/
Overriding model.yaml nc=80 with nc=3

from n params module arguments
0 -1 1 928 ultralytics.nn.modules.conv.Conv [3, 32, 3, 2]
1 -1 1 18560 ultralytics.nn.modules.conv.Conv [32, 64, 3, 2]
2 -1 1 26080 ultralytics.nn.modules.block.C3k2 [64, 128, 1, False, 0.25]
3 -1 1 147712 ultralytics.nn.modules.conv.Conv [128, 128, 3, 2]
4 -1 1 103360 ultralytics.nn.modules.block.C3k2 [128, 256, 1, False, 0.25]
5 -1 1 590336 ultralytics.nn.modules.conv.Conv [256, 256, 3, 2]
6 -1 1 346112 ultralytics.nn.modules.block.C3k2 [256, 256, 1, True]
7 -1 1 1180672 ultralytics.nn.modules.conv.Conv [256, 512, 3, 2]
8 -1 1 1380352 ultralytics.nn.modules.block.C3k2 [512, 512, 1, True]
9 -1 1 656896 ultralytics.nn.modules.block.SPPF [512, 512, 5]
10 -1 1 990976 ultralytics.nn.modules.block.C2PSA [512, 512, 1]
11 -1 1 0 torch.nn.modules.upsampling.Upsample [None, 2, 'nearest']
12 [-1, 6] 1 0 ultralytics.nn.modules.conv.Concat [1]
13 -1 1 443776 ultralytics.nn.modules.block.C3k2 [768, 256, 1, False]
14 -1 1 0 torch.nn.modules.upsampling.Upsample [None, 2, 'nearest']
15 [-1, 4] 1 0 ultralytics.nn.modules.conv.Concat [1]
```

4. Input Processing (Image/Video/Webcam)

The trained model accepts multiple types of input sources, including:

- Static images
- Image folders
- Video files
- Live webcam feed

```
libpng warning: icCP: profile 'icc': 0h: PCS illuminant is not D50
Plotting labels to /content/runs/detect/train/labels.jpg...
libpng warning: icCP: profile 'icc': 0h: PCS illuminant is not D50
libpng warning: icCP: profile 'icc': 0h: PCS illuminant is not D50
libpng warning: icCP: profile 'icc': 0h: PCS illuminant is not D50
Image sizes 640 train, 640 val
Using 2 dataloader workers
Logging results to /content/runs/detect/train
Starting training for 60 epochs...
```

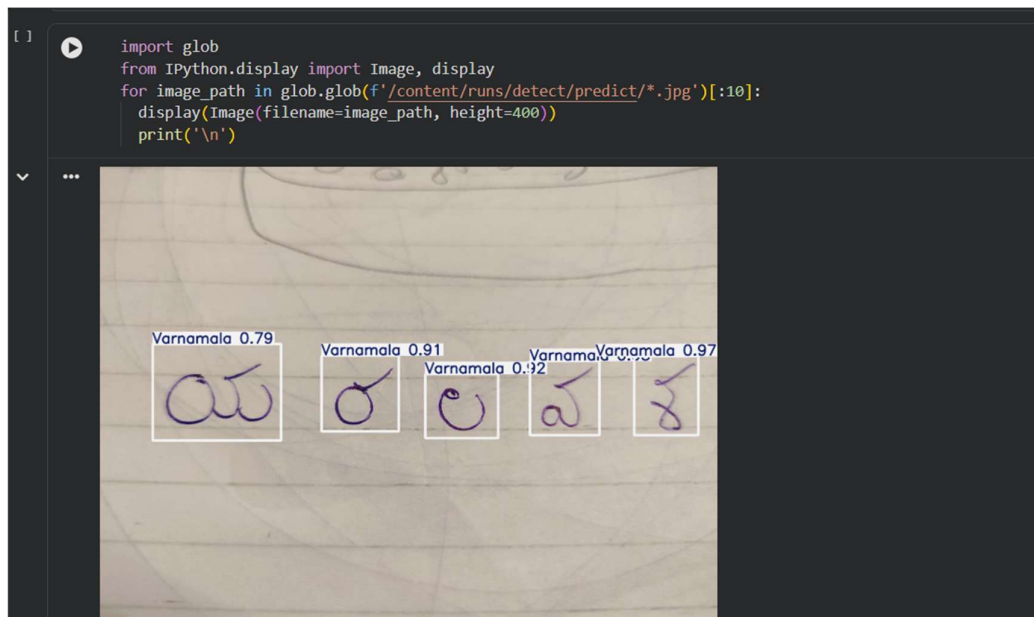
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
1/60	7.14G	3.253	3.612	1.477	4533	640: 100%	1/1 11.2s/it 11.2s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
	all	2	11	0.00425	0.159	0.00487	0.00238 1/1 4.5s/it 4.5s

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
2/60	7.14G	3.041	3.487	1.683	2099	640: 100%	1/1 2.4it/s 0.4s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
	all	2	11	0.00344	0.0476	0.00469	0.00234 1/1 7.5it/s 0.1s

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
3/60	7.14G	3.162	3.597	1.436	3912	640: 100%	1/1 2.0it/s 0.5s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
	all	2	11	0.00344	0.0476	0.00518	0.00259 1/1 12.4it/s 0.1s

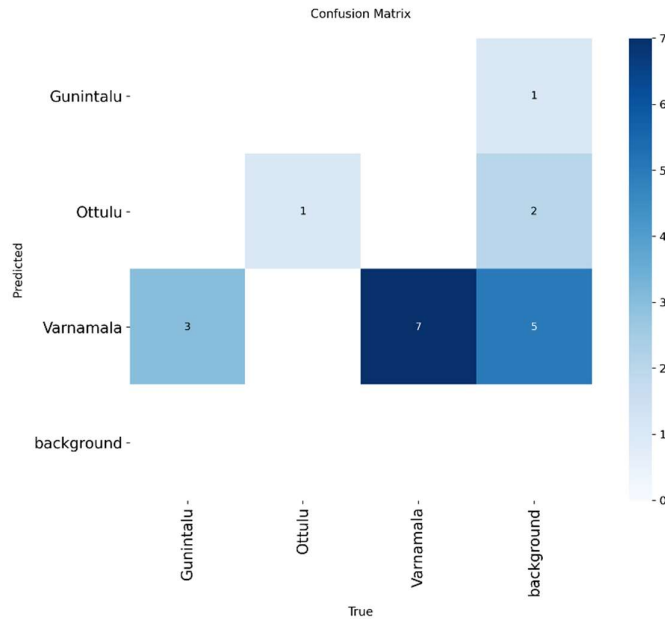
Input data is preprocessed (such as resizing and formatting) to match the model requirements before being passed for detection.

5. Detection and Segmentation

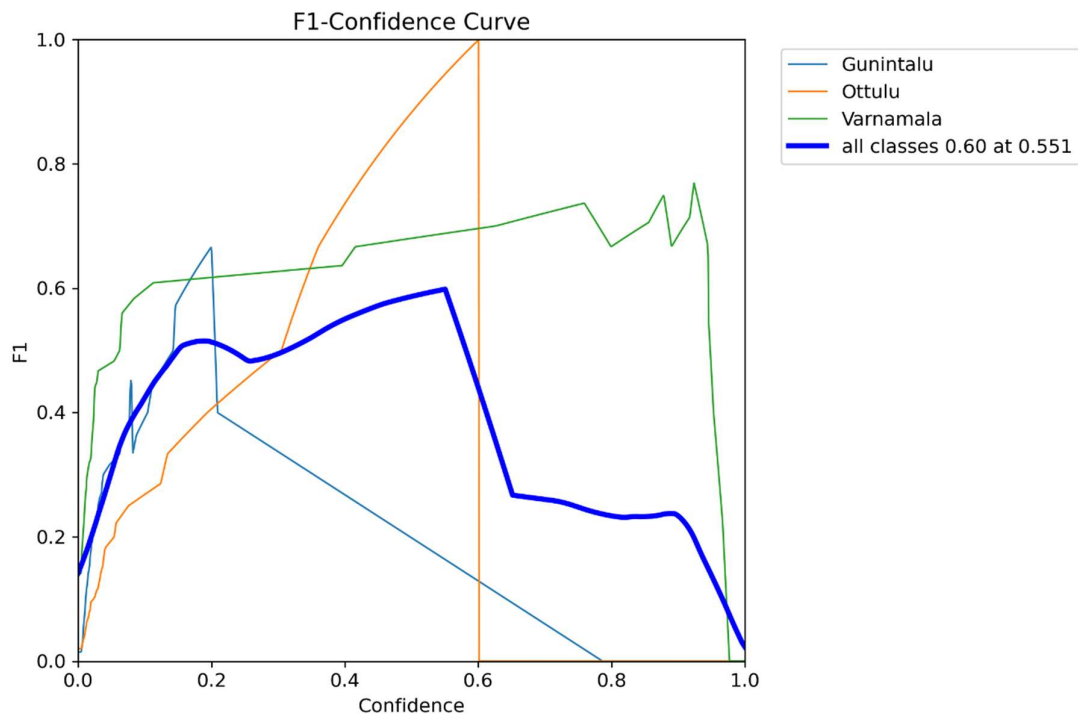


The YOLOv11 model processes the input data and detects Telugu characters by generating bounding boxes around them. Each detected object is assigned:

- A class label (character type)



- A confidence score

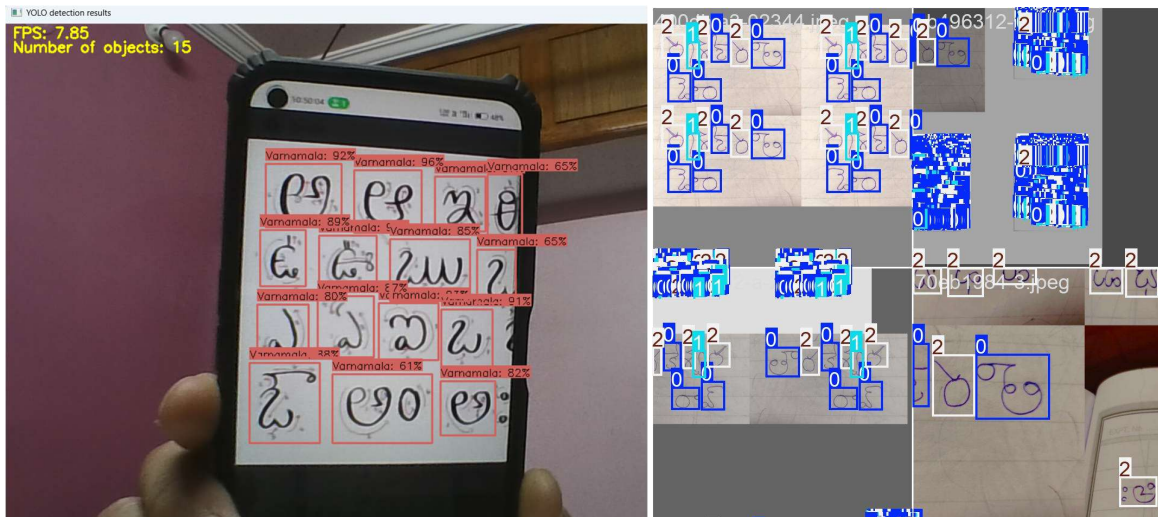


Segmentation is achieved by isolating characters based on these bounding boxes.

6. Output Visualization (Bounding Boxes & Labels)

The final output displays the detected characters with bounding boxes, labels, and confidence scores. Additional features include:

- Real-time visualization
- Object counting
- FPS (frames per second) display
- Option to save or record output



8. Tools/Technologies Used

- **Programming Language:** Python
- **Model:** YOLOv11 (Ultralytics)
- **Annotation Tool:** Label Studio
- **Environment:** Anaconda
- **Libraries:** OpenCV, NumPy
- **Framework:** Ultralytics YOLO

Details of various Tools/Technologies used and their significance in the project

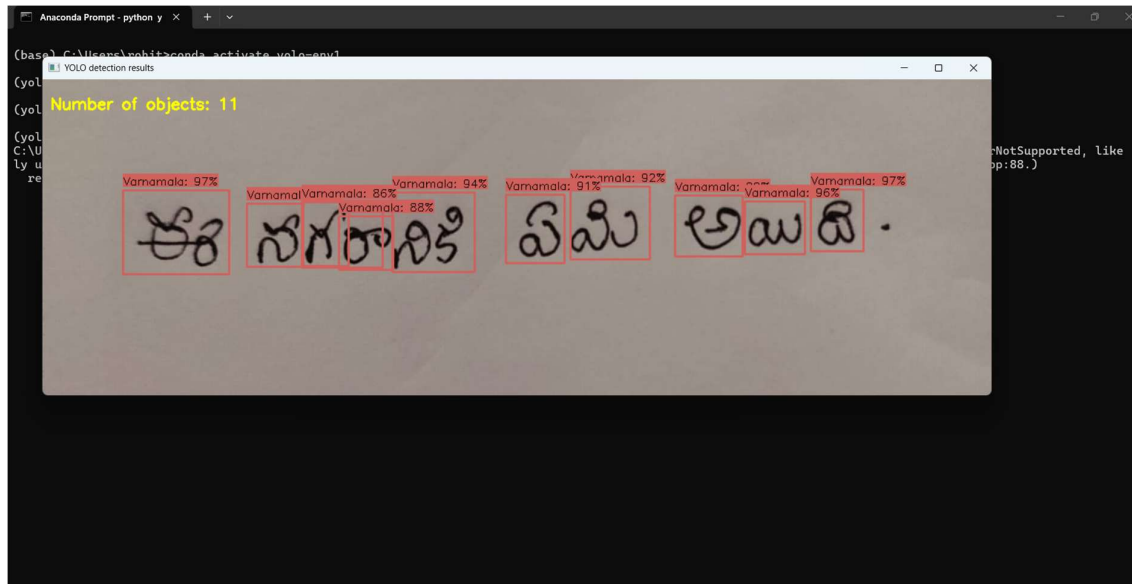
Implementation & Results

The system is implemented using Python and utilizes the Ultralytics YOLO framework for object detection. The model is loaded and used to perform inference on different input sources such as images, folders, videos, and live camera streams .

Key Features from Implementation

- Dynamic input handling (image, video, webcam)
- Bounding box detection with confidence threshold
- Real-time FPS calculation
- Object counting
- Option to record output video

Results



- Successful segmentation of Telugu handwritten characters
- Real-time detection capability
- Efficient performance across multiple input sources

9. Conclusion

This project demonstrates an effective approach for Telugu handwritten character segmentation using YOLOv11. The use of Label Studio for annotation and Anaconda for environment setup ensures a smooth and scalable development process.

The system achieves accurate and real-time detection, making it suitable for applications such as document digitization and handwriting analysis. Future improvements can further enhance accuracy and extend the system to full character recognition.

10. References

- [1] Ultralytics, “YOLOv5/YOLOv11 Documentation,” *Ultralytics*, 2024. [Online]. Available: <https://docs.ultralytics.com/>
- [2] Label Studio, “Label Studio Documentation,” *Heartex*, 2024. [Online]. Available: <https://labelstud.io/>
- [3] Anaconda, “Anaconda Documentation,” *Anaconda Inc.*, 2024. [Online]. Available: <https://docs.anaconda.com/>
- [4] G. Bradski, “The OpenCV Library,” *Dr. Dobb’s Journal of Software Tools*, 2000.
- [5] S. van der Walt, S. C. Colbert, and G. Varoquaux, “The NumPy Array: A Structure for Efficient Numerical Computation,” *Computing in Science & Engineering*, vol. 13, no. 2, pp. 22–30, 2011.

PROJECT PROFORMA

Classification of the Project	Application	Product	Research	Review
	✓			

Programme Outcomes (POs) and Programme Specific Outcomes (PSOs):

At the end of the Computer Science and Engineering programme, students are expected to achieve the following outcomes. In the context of this project on Telugu Handwritten Character Segmentation using YOLOv11, these outcomes are addressed as follows:

Programme Outcomes (POs)

➤ **PO1 – Engineering Knowledge:**

The project applies knowledge of mathematics, computer vision, and deep learning concepts to solve the problem of handwritten character segmentation.

➤ **PO2 – Problem Analysis:**

The problem of segmenting complex Telugu handwritten characters is analyzed by studying script structure, challenges, and existing limitations.

➤ **PO3 – Design/Development of Solutions:**

A deep learning-based solution using YOLOv11 is designed to accurately detect and segment characters.

➤ **PO4 – Conduct Investigations:**

The project involves dataset preparation, experimentation, and evaluation using metrics such as precision, recall, and mAP.

➤ **PO5 – Modern Tool Usage:**

Modern tools like Python, YOLOv11, OpenCV, NumPy, and Label Studio are used for implementation and analysis.

➤ **PO6.– The Engineer and Society:**

The project contributes to society by enabling document digitization and supporting regional language processing.

➤ **PO7 – Environment and Sustainability:**

Digitization reduces paper usage and promotes sustainable data management.

➤ **PO8 – Ethics:**

Ethical practices are followed in dataset creation and usage of technology.

➤ **PO9 – Individual and Team Work:**

The project is developed collaboratively by a team, ensuring effective coordination.

➤ **PO10 – Communication:**

Results are documented through reports and presentations.

➤ **PO11 – Project Management and Finance:**

Project planning, execution, and time management are demonstrated.

➤ **PO12 – Life-long Learning:**

The project encourages learning of advanced technologies like deep learning and computer vision.

Programme Specific Outcomes (PSOs)

➤ **PSO1:**

Application of programming skills using Python for developing the segmentation system.

➤ **PSO2:**

Development of an AI-based application using deep learning techniques.

➤ **PSO3:**

Ability to apply computer science knowledge ethically for solving real-world problems.

Project Course Outcomes

- **CO1:** Formulated the problem of Telugu handwritten character segmentation with proper assumptions.

- **CO2:** Conducted literature review on traditional and deep learning-based approaches.
- **CO3:** Designed and implemented the YOLOv11-based segmentation system.
- **CO4:** Evaluated performance using metrics and analyzed results.
- **CO5:** Documented the project in the form of a technical report and presentation.

CO–PO Mapping (Analysis)

The project strongly supports multiple Programme Outcomes:

- High mapping with **PO1 to PO5** as the project involves technical knowledge, problem-solving, design, experimentation, and tool usage.
- Strong alignment with **PO9 and PO10** due to teamwork and communication.
- Moderate contribution to **PO6, PO7, and PO8** through societal impact, sustainability, and ethics.
- Fully supports **PSO1, PSO2, and PSO3** through programming, AI application, and real-world problem solving.

UN Sustainable Development Goals (UN-SDG) Mapping

Relevant SDG for the Project

- **SDG 4 – Quality Education:**
The project contributes to digital learning by enabling digitization of handwritten Telugu documents, making educational resources more accessible.
- **SDG 9 – Industry, Innovation, and Infrastructure:**
The use of advanced AI models like YOLOv11 promotes innovation in computer vision and language processing technologies.
- **SDG 11 – Sustainable Cities and Communities:**
Supports preservation and digital accessibility of regional language documents.

Project SDG Mapping Table

Project Title	Project Type	SDG Target	Justification
Telugu Handwritten Character Segmentation Using YOLOv11	Capstone Project	SDG 4 – Quality Education	Helps digitize handwritten Telugu documents for better accessibility
		SDG 9 – Industrial Innovation	Uses advanced AI (YOLOv11) for real-world problem solving
		SDG 11 – Sustainable Communities	Preserves regional language data digitally



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